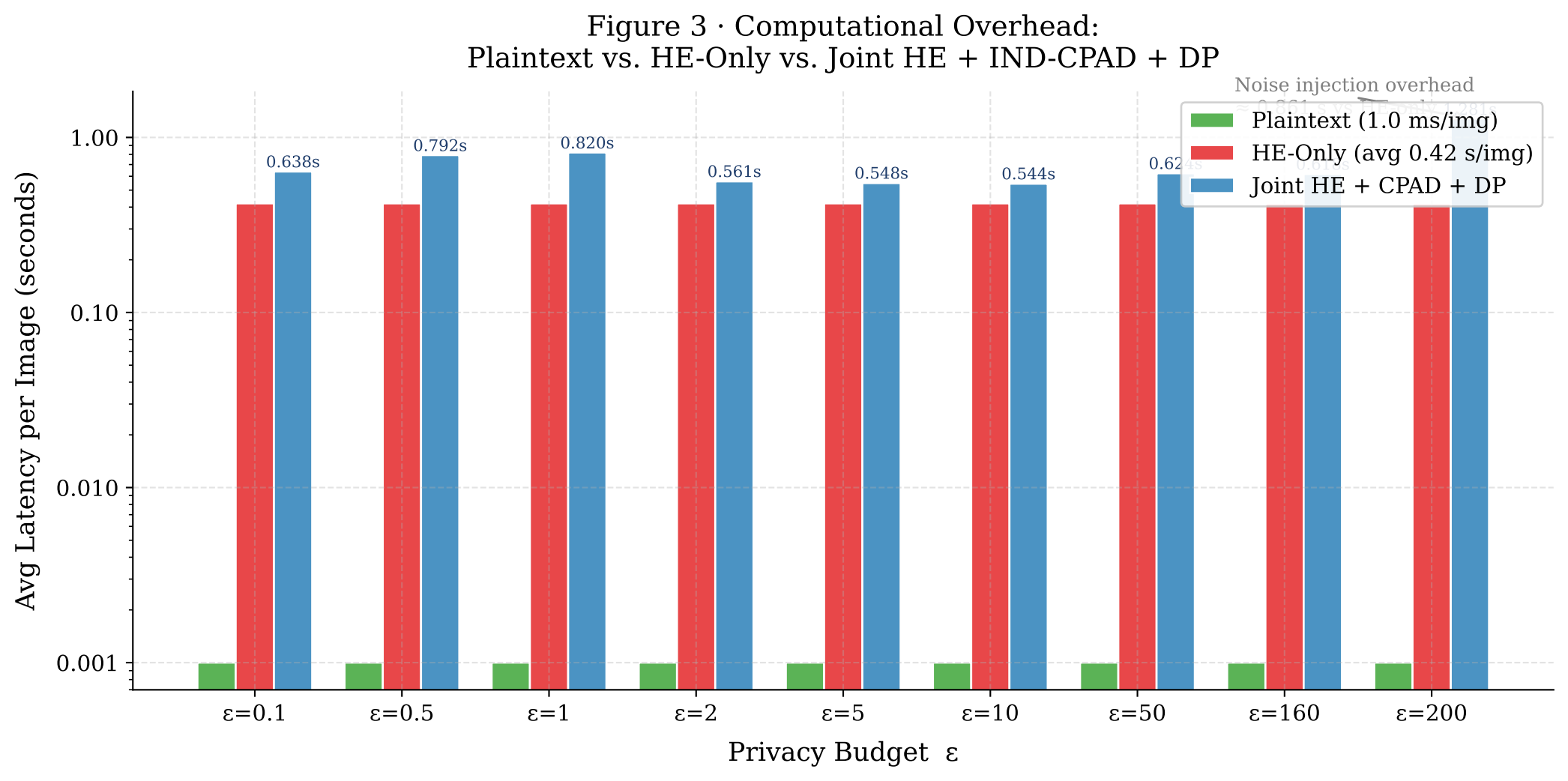
Figure 3 · Computational Overhead:
Plaintext vs. HE-Only vs. Joint HE + IND-CPAD + DP
Noise injection overhead
Plaintext (1.0 ms/img)
HE-Only (avg 0.42 s/img)
Joint HE + CPAD + DP
Avg Latency per Image (seconds)
1.00
0.10
0.010
0.001
0.638s
0.792s
0.820s
0.561s
0.548s
0.544s
ε=0.1
ε=0.5
ε=1
ε=2
ε=5
ε=10
ε=50
ε=160
ε=200
Privacy Budget ε